

Chapter 5

Energy Function Theory and Direct Methods

5.1 INTRODUCTION

Power system stability analysis is concerned with whether the postfault trajectory will settle to an acceptable operating condition. Direct methods for stability analysis use algorithmic procedures to assess, without integrating the postfault system, the stability property of the postfault trajectory by comparing the system energy at the initial state of the postfault trajectory to a critical energy value. Direct methods not only avoid the time-consuming procedure of numerical integration of the postfault system, but they also provide a quantitative measure of the degree of system stability. Given a power system transient stability model with specified fault-on systems and a specified postfault system, direct methods for transient stability analysis are composed of the following steps:

- Step 1.* Numerically simulate the fault-on trajectory.
- Step 2.* Compute the initial point of the postfault system.
- Step 3.* Construct an energy function for the postfault power system.
- Step 4.* Compute the energy function value at the initial point of the postfault system.
- Step 5.* Compute the critical energy for the fault-on trajectory.
- Step 6.* Perform direct stability analysis by comparing the system energy at the initial state of the postfault system (computed at Step 4) with the critical energy (computed at Step 5). If the former is smaller than the latter, then the postfault trajectory will be stable; otherwise, it may be unstable.

Steps 1 and 2 require efficient numerical methods for simulating the fault-on trajectory. The construction of an energy function for a postfault power system (needed in Step 3) will be discussed in the next two chapters. Step 4 is a direct

calculation of the energy function value at the initial condition obtained in Step 2. In Step 6, direct methods directly determine whether a postfault power system will remain stable by comparing the system energy computed in Step 4 to the critical energy computed in Step 5. Hence, it is crucial that an energy function capturing the system energy value be used to compute the system energy and an accurate critical energy value (for the fault-on trajectory) be calculated in direct methods. The topic of how to construct an energy function for a postfault power system will be discussed in this chapter and in the next chapter. The various methods for computing critical energy values along with the theoretical foundation of these methods will be presented in later chapters.

The theoretical basis of direct methods for the stability assessment of a postfault power system is the knowledge of the stability region of the postfault power system; if the initial condition of the postfault system lies inside the stability region of a desired postfault stable equilibrium point (SEP), then one can ensure, without performing numerical integrations, that the ensuing postfault trajectory will converge to the desired point. Therefore, knowledge of a stability region plays an important role in the theoretical foundation of direct methods.

In this chapter, a comprehensive energy function theory for general nonlinear autonomous dynamical systems will be presented. In addition, the theoretical basis for using energy functions to estimate the stability regions of large-scale nonlinear systems will be presented. The energy function theory has been applied to power system transient stability models to develop a theoretical foundation for direct methods. We will present this development in later chapters.

5.2 ENERGY FUNCTIONS

In this section, we study energy functions that can be viewed as extension of the Lyapunov functions. Energy functions are useful for global analysis of system trajectories and for estimating stability regions and quasi-stability regions, among others. The energy function theory for general nonlinear autonomous dynamical systems will be presented in the next section.

We consider a general nonlinear autonomous dynamical system described by the following equation:

$$\dot{x}(t) = f(x(t)). \quad (5.1)$$

We say a function, $V: \mathbb{R}^n \rightarrow \mathbb{R}$, is an energy function for the system (Eq. 5.1) if the following three conditions are satisfied:

- (i) The derivative of the energy function $V(x)$ along any system trajectory, $x(t)$, is nonpositive; that is,

$$\dot{V}(x(t)) \leq 0.$$

- (ii) If $x(t)$ is a nontrivial trajectory (i.e., $x(t)$ is not an equilibrium point), then along the nontrivial trajectory $x(t)$, the set

$$\{t \in R: \dot{V}(x(t)) = 0\}$$

has a measure zero in R .

- (iii) That a trajectory $x(t)$ has a bounded value of $V(x(t))$ for $t \in R^+$ implies that the trajectory $x(t)$ is also bounded. Stating this in brief, that $V(x(t))$ is bounded, implies that $x(t)$ itself is also bounded.

Property 1 indicates that the energy function is nonincreasing along its trajectory but does not imply that the energy function is strictly decreasing along its trajectory. There may exist a time interval $[t_1, t_2]$ such that $\dot{V}(x(t)) = 0$ for $t \in [t_1, t_2]$. Properties 1 and 2 imply that the energy function is strictly decreasing along any system trajectory. Property 3 states that the energy function is a proper map along any system trajectory but need not be a proper map for the entire state space. Recall that a proper map is a function $f: X \rightarrow Y$ such that for each compact set $D \in Y$, the set $f^{-1}(D)$ is compact in X . It will become clear that property 3 can be viewed as a “dynamic” proper map. From the above definition of the energy function, it is obvious that an energy function may not be a Lyapunov function.

As an illustration of the energy function, we consider the following classical transient stability model and derive an energy function for the model. Consider a power system consisting of n generators. Let the loads be modeled as constant impedances. Under the assumption that the transfer conductance of the reduced network after eliminating all load buses is zero, the dynamics of the i th generator can be represented by the equations

$$\begin{aligned} \dot{\delta}_i &= \omega_i \\ M_i \dot{\omega}_i &= P_i - D_i \omega_i - \sum_{\substack{j=1 \\ j \neq i}}^{n+1} V_i V_j B_{ij} \sin(\delta_i - \delta_j), \end{aligned} \quad (5.2)$$

where the voltage at node $i + 1$ is served as the reference; that is, $\delta_{i+1} = 0$. The set of equations (Eq. 5.2) for $i = 1, 2, \dots, n$ describes the power system and can be written in matrix form:

$$\begin{aligned} \dot{\delta} &= \omega \\ M \dot{\omega} &= P - D\omega - f(\delta, V), \end{aligned} \quad (5.3)$$

where $\delta = (\delta_1, \dots, \delta_n)^T$, $\omega = (\omega_1, \dots, \omega_n)^T$, $P = (P_1, \dots, P_n)^T$, $V = (V_1, \dots, V_n)^T$, and $f = (f_1, \dots, f_n)^T$ are n -vectors; $M = \text{diag}(M_1, \dots, M_n)$, and $D = \text{diag}(D_1, \dots, D_n)$ are $n \times n$ positive diagonal matrices; and

$$f_i(\delta, V) = \sum_{\substack{j=1 \\ j \neq i}}^{n+1} V_i V_j B_{ij} \sin(\delta_i - \delta_j).$$

The equations for the prefault, fault-on, and postfault systems have the same form as Equation 5.2 except that the B_{ij} is different due to the change in network topology. This is a version of the so-called *classical model* of the power system. We

next show that there exists an energy function, $V(\delta, \omega)$, for the classical model (Eq. 5.3).

Consider the following function:

$$V(\delta, \omega) = \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2 - \sum_{i=1}^n P_i (\delta_i - \delta_i^s) - \sum_{i=1}^n \sum_{j=i+1}^{n+1} V_i V_j B_{ij} \{ \cos(\delta_i - \delta_j) - \cos(\delta_i^s - \delta_j^s) \}, \quad (5.4)$$

where $x^s = (\delta^s, 0)$ is the SEP under consideration.

Differentiating V along the trajectory $(\delta(t), \omega(t))$ of Equation 5.2 gives

$$\dot{V}(\delta(t), \omega(t)) = \sum_{i=1}^n \left(\frac{\partial V}{\partial \delta_i} \dot{\delta}_i + \frac{\partial V}{\partial \omega_i} \dot{\omega}_i \right) = - \sum_{i=S_n} D_i \omega_i^2 \leq 0. \quad (5.5)$$

For the trajectory starting at an equilibrium point, $(\delta^s, 0)$, we have $\omega(t) = 0$; hence,

$$\dot{V}(\delta(t), \omega(t)) = 0. \quad (5.6)$$

Suppose that there is an interval, $t \in [t_1, t_2]$, such that

$$\dot{V}(\delta(t), \omega(t)) = 0 \quad t \in [t_1, t_2]. \quad (5.7)$$

Hence,

$$\omega(t) = 0 \quad t \in [t_1, t_2]. \quad (5.8)$$

But this implies $\omega(t) = 0$ and $\delta(t) = \text{constant}$ for $t \in [t_1, t_2]$. It then follows from Equation 5.2 that

$$P_i - \sum_{\substack{j \neq i \\ j=1}}^{n+1} V_i V_j B_{ij} \sin(\delta_i - \delta_j) = 0, \quad (5.9)$$

which are precisely the equations for the equilibrium point of Equation 5.2. Therefore, $(\delta(t), \omega(t))$ $t \in [t_1, t_2]$ must be on an equilibrium point.

Let us first integrate Equation 5.2 for ω_i :

$$\omega_i(t) = e^{-D_i/M_i t} \omega_i(0) + \int_0^t e^{-D_i/M_i(t-s)} \left\{ P_i - \sum_{\substack{j \neq i \\ j=1}}^{n+1} V_i V_j B_{ij} \sin(\delta_i(s) - \delta_j(s)) \right\} ds. \quad (5.10)$$

The term in the bracket is uniformly bounded, say, by a_i ; that is,

$$\left| P_i - \sum_{\substack{j \neq i \\ j=1}}^{n+1} V_i V_j B_{ij} \sin(\delta_i(s) - \delta_j(s)) \right| \leq a_i. \quad (5.11)$$

Since D_i and M_i are positive numbers, we have from Equation 5.10

$$|\omega_i(t)| \leq |\omega_i(0)| + a_i \frac{M_i}{D_i}; \quad (5.12)$$

that is, $\omega_i(t)$ is bounded by $b_i := |\omega_i(0)| + a_i M_i / D_i$.

We next show that condition (iii) of the energy function is satisfied, suppose $V(\delta(t), \omega(t))$ is bounded below and above, say, by c_1 and c_2 , respectively. Then, we have

$$c_1 - \frac{1}{2} \sum_{i=1}^n M_i b_i^2 < - \sum_{i=1}^n P_i (\delta_i - \delta_i^s) - \sum_{i=1}^n \sum_{j=i-1}^{n+1} V_i V_j B_{ij} \{ \cos(\delta_i - \delta_j) - \cos(\delta_i^s - \delta_j^s) \} < c_2. \quad (5.13)$$

But the second term on the right-hand side is uniformly bounded, say, by c ; that is,

$$\left| \sum_{i=1}^n \sum_{j=i-1}^{n+1} V_i V_j B_{ij} \{ \cos(\delta_i - \delta_j) - \cos(\delta_i^s - \delta_j^s) \} \right| < c. \quad (5.14)$$

Substituting Equation 5.14 into Equation 5.13, we get

$$c_1 - \frac{1}{2} \sum_{i=1}^n M_i b_i^2 - c < - \sum_{i=1}^n P_i (\delta_i - \delta_i^s) < c_2 + c. \quad (5.15)$$

Hence, the term $P^T \delta$ is bounded. It is shown in Arapostathis et al. (1982) that $P^T \delta$ being bounded implies $\delta(t)$ is bounded. This result, together with Equation 5.12, asserts that the trajectory $((\delta(t), \omega(t)))$ on the stability boundary is bounded. Hence, condition (iii) of the energy function is satisfied.

5.3 ENERGY FUNCTION THEORY

The theoretical basis of direct methods for power system transient stability assessment is the knowledge of the stability region. We present analytical results showing that the stability regions of a class of nonlinear systems can be completely characterized. For this class of nonlinear systems, which have energy functions, a complete characterization of the stability boundary will be derived and optimal schemes to estimate its stability regions via an energy function will be presented.

We first examine the behaviors of system trajectories of nonlinear systems having energy functions. In general, the dynamic behaviors of trajectories of general nonlinear systems can be very complicated. The asymptotic behaviors (i.e., the ω -limit set) of trajectories can be quasi-periodic trajectories or chaotic trajectories. However, as shown below, if the underlying dynamical system has some special properties, then the system may admit only simple trajectories. For instance, every trajectory of the system (Eq. 5.1) having an energy function has only two modes of behaviors: its trajectory either converges to an equilibrium point or goes to infinity (becomes unbounded) as time increases or decreases. This result is explained in the following theorem.

Theorem 5.1: Global Behavior of Trajectories

If there exists a function satisfying conditions (i) and (ii) of the energy function for the system (Eq. 5.1), then every bounded trajectory of the system (Eq. 5.1) converges to one of the equilibrium points.

Theorem 5.1 asserts that there does not exist any limit cycle (oscillation behavior) or bounded complicated behavior such as an almost periodic trajectory and a chaotic motion in the system. Applying this result to power system models, it indicates that for a power system model having an energy function, there is no complicated behavior such as closed orbits (limit cycle), quasi-periodic trajectories, and chaotic motions in the state space of the model.

We next show a sharper result, asserting that every trajectory on the stability boundary must converge to one of the unstable equilibrium points (UEPs) on the stability boundary.

Theorem 5.2: Trajectories on the Stability Boundary

If there exists an energy function for the system (Eq. 5.1), then every trajectory on the stability boundary $\partial A(x_s)$ converges to one of the equilibrium points on the stability boundary $\partial A(x_s)$.

The significance of this theorem is that it offers an effective way to characterize the stability boundary. In fact, Theorem 5.2 asserts that the stability boundary $\partial A(x_s)$ is contained in the union of stable manifolds of the UEPs on the stability boundary. One corollary of Theorem 5.2 shown next will be extended to develop a theoretical foundation for direct methods, in particular, the closest UEP method and the controlling UEP method.

Corollary 5.3: Energy Function and Stability Boundary

If there exists an energy function for the system (Eq. 5.1), which has an asymptotically stable equilibrium point, x_s (but not globally asymptotically stable), then the stability boundary $\partial A(x_s)$ is contained in the set, which is the union of the stable manifolds of the UEPs on the stability boundary $\partial A(x_s)$; that is,

$$\partial A(x_s) \subseteq \bigcup_{x_i \in \{E \cap \partial A(x_s)\}} W^s(x_i).$$

The following two theorems give interesting results on the structure of the equilibrium points on the stability boundary. Moreover, it presents a necessary condition for the existence of certain types of equilibrium points on a *bounded* stability boundary. The proof is similar to that of Theorem 3.16 and is therefore omitted.

Theorem 5.4: Structure of Equilibrium Points on the Stability Boundary

If there exists an energy function for the system (Eq. 5.1), which has an asymptotically stable equilibrium point, x_s (but not globally asymptotically stable), then the stability boundary $\partial A(x_s)$ must contain at least one type-one equilibrium point. If, furthermore, the stability region is bounded, then the stability boundary $\partial A(x_s)$ must contain at least one type-one equilibrium point and one source.

The contrapositive of Theorem 5.4 leads to the following corollary, which is useful in predicting the unboundedness of a stability region.

Theorem 5.5: Sufficient Condition for Unbounded Stability Region

If there exists an energy function for the system (Eq. 5.1), which has an asymptotically stable equilibrium point, x_s (but not globally asymptotically stable), and if $\partial A(x_s)$ contains no source, then the stability region $A(x_s)$ is unbounded.

We next address the issue of whether it is possible that two equilibrium points of the system (Eq. 5.1) have the same energy function value $V(\cdot)$ (i.e., $V(x_1) = V(x_2)$ for $x_1, x_2 \in E$). In the context of direct methods, this issue is related to the uniqueness of the closest UEP and the uniqueness of the controlling UEP. Note that all the potential energy $V_p(x)$ in the (total) energy functions $V(\cdot)$ for existing power system transient stability models have the property that $\nabla V_p(x) = 0$ at any equilibrium point. To answer this question, we first observe the conditions for two equilibrium points x_1 and x_2 to have the relationships that $V_p(x_1) = V_p(x_2)$ and $\nabla V_p(x_1) = \nabla V_p(x_2)$. This is a set of $(2n + 1)$ nonlinear algebraic equations with $2n$ unknowns (x_1, x_2); it is unlikely that there exists a solution. A rigorous justification of this observation is presented in Theorem 5.6. This theorem states that generically, all the equilibrium points of the system (Eq. 5.1) have distinct energy function values. We recall the concept of a generic property. Let X be a complete metric space and let $P(x)$ be a statement about points x in X . We say that $P(x)$ is a *generic property* if the set of points where it holds true contains a countable intersection of open dense sets.

The following result can be proved via Thom's famous transversality theorem (Hirsch, 1976; Munkres, 1975).

Theorem 5.6: Energy Functions and Equilibrium Points

Let $V(\cdot)$ be an energy function for the system (Eq. 5.1) and $\nabla V(x) = 0$ at the equilibrium points of the system (Eq. 5.1). Then, the property that all the equilibrium points of the system (Eq. 5.1) have distinct values of energy function $V(\cdot)$ is generic.

We next apply the analytical results derived in this section to the classical transient stability model (Eq. 5.2), which admits an energy function for the model. First, the hyperbolicity of equilibrium points of the classical power system stability model (Eq. 5.3) is shown.

Proposition 5.7: Hyperbolic Equilibrium Points

Consider the classical power system stability model (Eq. 5.3). If the Jacobian matrix $\left(\frac{\partial f}{\partial \delta}\right)$ is nonsingular at an equilibrium point, then the equilibrium point is hyperbolic.

Proof: Let $\hat{x} = (\hat{\delta}, \hat{w})$ be an equilibrium point of Equation 5.3. The Jacobian at $\hat{x}$ is given by

$$J(\hat{x}) = \begin{bmatrix} 0 & I \\ -M^{-1}F(\hat{\delta}) & -M^{-1}D \end{bmatrix}, \quad (5.16)$$

where the ij th element of the matrix $F(\hat{\delta})$ is $\partial f_i / \partial \delta_j(\hat{\delta})$. Let λ be an eigenvalue of $J(\hat{x})$ and $x = (x_1, x_2)$ be a corresponding eigenvector:

$$\begin{bmatrix} 0 & I \\ -M^{-1}F(\hat{\delta}) & -M^{-1}D \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (5.17)$$

or

$$x_2 = \lambda x_1 \text{ and} \quad (5.18)$$

$$-M^{-1}F(\hat{\delta})x_1 - M^{-1}Dx_2 = \lambda x_2. \quad (5.19)$$

Substituting Equations 5.18 into Equation 5.19 and premultiplying by $x_1^T M$, we get

$$\lambda^2 x_1^T M x_1 + \lambda x_1^T D x_1 + x_1^T F(\hat{\delta}) x_1 = 0. \quad (5.20)$$

We are going to show that $\lambda \neq 0$ and $\lambda \neq j\alpha$; hence, the equilibrium point is hyperbolic. The two cases are considered separately.

1. Suppose $\lambda = 0$; then, Equation 5.20 becomes

$$x_1^T F(\hat{\delta}) x_1 = 0,$$

which contradicts the assumption that $F(\hat{\delta})$ is nonsingular.

2. Suppose $\lambda = j\alpha$; then, Equation 5.20 becomes

$$\alpha^2 x_1^T M x_1 = x_1^T F(\hat{\delta}) x_1 \text{ and} \quad (5.21)$$

$$j\alpha x_1^T D x_1 = 0. \quad (5.22)$$

Again this is a contradiction because D is assumed to be nonsingular. This completes the proof.

The hyperbolicity of the equilibrium points of the classical power system stability model (Eq. 5.3) implies that every equilibrium point is isolated. We next apply Theorem 5.2 to conclude that every trajectory on the stability boundary of the classical power system stability model (Eq. 5.3) converges to one of the equilibrium points, and the following characterization of the stability boundary for the classical power system stability model (Eq. 5.3) is derived.

Proposition 5.8: Characterization of Stability Boundary

Let x_s be an asymptotically stable equilibrium point of the classical power system stability model (Eq. 5.3), then the stability boundary $\partial A(x_s)$ is contained in the set, which is the union of the stable manifolds of the UEPs on the stability boundary $\partial A(x_s)$; that is,

$$\partial A(x_s) \subseteq \bigcup_{x_i \in \{E \cap \partial A(x_s)\}} W^s(x_i).$$

The stability region of the classical power system stability model (Eq. 5.3) is shown to be unbounded as stated in the following.

Proposition 5.9: Unbounded Stability Region

The stability boundary $\partial A(x_s)$ of an asymptotically stable equilibrium point of the classical power system stability model (Eq. 5.3) is unbounded.

Proof: We will show that the equilibrium points of Equation 5.3 do not contain a source. Specifically, we will show that the Jacobian $J(\hat{x})$ always has eigenvalues with negative real parts. This is done by the application of the inertia theorem (Wimmer, 1974), which states that if H is a nonsingular Hermitian matrix and A has no eigenvalues on the imaginary axis (i.e., $n_c(A) = n_c(H) = 0$), then $AH + HA^* \geq 0$ implies that

$$\text{In}(A) = \text{In}(H). \quad (5.23)$$

Now we choose H to be the symmetric matrix:

$$H = \begin{bmatrix} -F(\hat{x})^{-1} & 0 \\ 0 & -M^{-1} \end{bmatrix}. \quad (5.24)$$

We have

$$J(\hat{x})H + HJ(\hat{x})^T = 2 \begin{bmatrix} 0 & 0 \\ 0 & M^{-1}DM^{-1} \end{bmatrix}. \quad (5.25)$$

Hence,

$$\ln(J(\hat{x})) = \ln \begin{bmatrix} -F(\hat{x})^{-1} & 0 \\ 0 & -M^{-1} \end{bmatrix}. \quad (5.26)$$

Since M is a diagonal matrix with positive elements, Equation 5.26 implies that, at any equilibrium point $\hat{x}$, the Jacobian $J(\hat{x})$ has at least n eigenvalues with negative real parts. This completes the proof.

Remark: It should be emphasized that Proposition 5.9 asserts that the stability region is unbounded for the power system model under consideration. In practice, of course, constraints from devices not modeled here would prevent the actual system from operating with a large deviation of frequency and angle differences.

5.4 ESTIMATING STABILITY REGION USING ENERGY FUNCTIONS

In this section, we focus on how to characterize, via an energy function, the stability region of a high-dimension nonlinear system such as a power system. This characterization will prove important in the direct methods. We consider the following set:

$$S_v(k) = \{x \in R^n : V(x) < k\}, \quad (5.27)$$

where $V(\cdot): R^n \rightarrow R$ is an energy function. Sometimes, we drop the subscript v of $S_v(k)$, simply writing $S(k)$, when it is clear from the context. We shall call the boundary of the set (Eq. 5.27) $\partial S(k) := \{x \in R^n : V(x) = k\}$ the *level set* (or *constant energy surface*) and k the *level value*. If k is a regular value (i.e., $\nabla V(x) \neq 0$, for all $x \in V^{-1}(k)$), then by the inverse function theorem, $\partial S(k)$ is a $C^r(n-1)$ -dimensional submanifold of R^n (Hurewicz and Wallman, 1948, p. 46). Moreover, if $r > n-1$, then by the Morse–Sard theorem, the set of regular values of V is residual; in other words, “almost all” level values are regular. In particular, for almost all values of k , the level set $\partial S(k)$ is a $C^r(n-1)$ -dimensional submanifold.

Generally speaking, this set $S(k)$ can be very complicated with several connected components even for the two-dimensional case. Let

$$S(k) = S^1(k) \cup S^2(k) \cup \dots \cup S^m(k), \quad (5.28)$$

where $S^i(k) \cap S^j(k) = \emptyset$ when $i \neq j$. That is, each of these components is connected and disjoint from each other. Since $V(\cdot)$ is continuous, $S(k)$ is an open set. Because $S(k)$ is an open set, the level set $\partial S(k)$ is of $(n-1)$ dimensions. Furthermore, each component of $S(k)$ is an invariant set.

In spite of the possibility that a constant energy surface may contain several disjoint connected components, there is an interesting relationship between the constant energy surface and the stability boundary. This relationship is that at most, one connected component of the constant energy surface $\partial S(r)$ has a nonempty intersection with the stability region $A(x_s)$. This relationship is established in Theorem 5.10.

Theorem 5.10: Constant Energy Surface and Stability Region

Let x_s be a SEP of the system (Eq. 5.1) and $A(x_s)$ be its stability region. Then, the set $S(r)$ contains only one connected component, which has a nonempty intersection with the stability region $A(x_s)$ if and only if $r > V(x_s)$.

Motivated by Theorem 5.10, we shall use the notation $S_{x_s}(r)$ to denote the connected set of $S(r)$ (whose level value is r) containing the SEP x_s . We drop the subscript x_s of $S_{x_s}(r)$ when it is clear from the context. In Figure 5.1, the relation between the constant energy surfaces at different level values and the stability region $A(x_s)$ is shown. It is observed from this figure that the connected set $S_{x_s}(r)$ with a level value r smaller than the critical value is very conservative in the approximation of the stability boundary $\partial A(x_s)$. As the set $S_{x_s}(r)$ is expanded by increasing the level value r , the approximation gets improved until this constant energy surface hits the stability boundary $\partial A(x_s)$ at some point. This point will be shown to be a UEP. We call this point the *closest UEP* of the SEP x_s with respect to the energy function $V(\cdot)$.

Furthermore, as we increase the level value r , the connected set $S_{x_s}(r)$ would contain points that lie outside the stability region $A(x_s)$. It is therefore inappropriate to approximate the stability region $A(x_s)$ by the connected set $S_{x_s}(r)$ with a level value higher than that of the closest UEP. From this figure, it becomes obvious that among the several connected sets of the constant energy surface, there is only one connected component of the constant energy surface that has a nonempty intersection with the stability region $A(x_s)$.

Topological and dynamic characterizations for the point on the stability boundary with the minimum value of an energy function will be derived. These characterizations are useful in identifying the point with the minimum value of an energy function over the stability boundary.

Theorem 5.11: Topological Characterization

Let x_s be a SEP and $A(x_s)$ be the stability region of the system (Eq. 5.1), which has an energy function. If the stability region $A(x_s)$ is not dense in R^n , then the point with the minimum value of the energy function over the stability boundary $\partial A(x_s)$ exists, and it must be a UEP.

Proof: By the conditions of energy functions, x_s is a SEP if and only if x_s is a local minimum of any energy function $V(x)$ associated with the system (Eq. 5.1). Hence, given any energy function $V(x)$, there exists a positive number n such that the set $\partial S(n)$, which is defined to be the connected component of the level set $\{x: V(x_s) \leq V(x) \leq n\}$ containing the SEP x_s , is closed and bounded.

Next, we investigate how the set $\partial S(n)$ changes as the level value n is varied. Suppose that n is a regular value of $V(\cdot)$; then, as a consequence of the implicit function theorem, the set $SC_n(x_s)$ is a manifold. Furthermore, as we increase the level value n to the new value, say, m with $m > n$, such that there are no critical

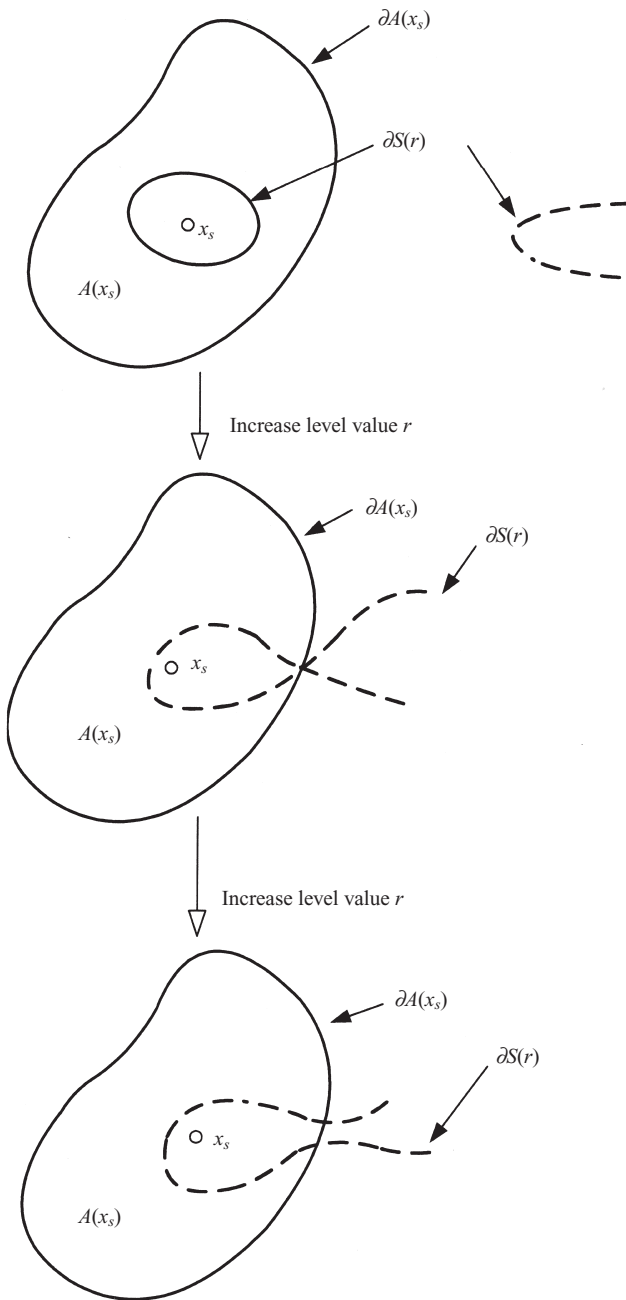


Figure 5.1 The relation between the constant energy surfaces $S(r)$ at different level values and the stability region $A(x_s)$.

points in the set $\{S(m) - S(n)\}$, then it can be shown (Franks, 1980) that the set $\partial S(m)$ is diffeomorphic to $\partial S(n)$. In particular, the set $\partial S(m)$ is closed and bounded.

Therefore, as we increase the level value n , the level set $\partial S(n)$ remains to be bounded until it hits a critical point, which is an equilibrium point of the system (Eq. 5.1). Now we argue that this equilibrium point, say, $\hat{x}$, must be lying on the stability boundary $\partial A(x_s)$. By contradiction, suppose that $V(\hat{x}) = \bar{m}$ and this equilibrium point $\hat{x} \in \overline{A(x_s)^c}$, here, $\overline{A(x_s)^c}$ denotes the complement of $\overline{A(x_s)}$. It is easy to see that the set $S\bar{m}$ has a nonempty intersection with the stability boundary $\partial A(x_s)$. Let $S := S(\bar{m}) \cap \partial A(x_s)$. We will show that S must contain at least an equilibrium point and thus contradicts the condition that $\hat{x}$ is the only equilibrium point lying in $S(\bar{m})$. We note the following:

1. $\partial A(x_s)$ is a closed and invariant set of the system (Eq. 5.1) and
2. $S_{\bar{m}}(x_s)$ is a compact and positive invariant set of the system (Eq. 5.1).

Hence, the set S is a compact, positive invariant set. Since any compact, positive invariant set must contain at least an ω -limit set and the ω -limit set of the system (Eq. 5.1) consists of equilibrium points, the set S must contain at least one equilibrium point, which is a contradiction, and we complete the proof.

Theorem 5.11 asserts that the point with the minimum value over the stability boundary must be a UEP. Theorem 5.12 gives a dynamic characterization of this UEP in terms of its unstable manifold. Note that this theorem holds without the transversality condition.

Theorem 5.12: Dynamic Characterization

Let x_s be a SEP and $A(x_s)$ be the stability region of the system (Eq. 5.1), which has an energy function. If the stability region $A(x_s)$ is not dense in R^n and $\hat{x}$ is the point with the minimum value of the energy function over the stability boundary $\partial A(x_s)$, then $W^u(\hat{x}) \cap A(\hat{x}) \neq \emptyset$.

Proof: We prove this theorem by contradiction. Suppose that the unstable manifold of $\hat{x}$ does not converge to the SEP x_s . Theorem 3.8 states that if x_s is a SEP and $\hat{x}$ is a hyperbolic equilibrium point, then $\hat{x}$ is on the stability boundary $\partial A(x_s)$ if and only if its unstable manifold has a nonempty intersection with the closure of the stability region; that is, $W^u(\hat{x}) \cap \overline{A(x_s)} \neq \emptyset$. Therefore, there exists a trajectory $x(t)$ in the unstable manifold $W^u(\hat{x})$ such that $x(t) \in \partial A(x_s)$ and $\lim_{t \rightarrow \infty} x(t) = \hat{x}$. In addition, the energy function value is strictly decreasing along any nontrivial trajectory of the system (Eq. 5.1). These two facts imply that there exist other points on the stability boundary with a lower energy function value than $V(\hat{x})$. This contradicts the fact that $\hat{x}$ possesses the lowest energy function value within the stability boundary $\partial A(x_s)$. This completes the proof.

We have analyzed some behaviors of the constant energy surface and the relationship between the constant energy surface and the stability region. The next

subject is approximating a stability region via a constant energy surface in an optimal way. Indeed, given an energy function, the central point in the energy function approach for estimating the stability region of a SEP is the determination of the critical level value for the energy function.

5.5 OPTIMAL SCHEMES FOR ESTIMATING STABILITY REGIONS

We discuss in this section how to optimally determine the critical level value of an energy function for estimating the stability boundary $\partial A(x_s)$. We use the notation $\bar{A}^c(x_s)$ to denote the complement of the set $\bar{A}_p(x_s)$. We use the notation $S_k(x_s)$ to denote the only component of the several disjoint connected components of S_k that contains the SEP x_s .

Theorem 5.13: Optimal Estimation

Consider the nonlinear system (Eq. 5.1) that has an energy function $V(x)$. Let x_s be an asymptotically stable equilibrium point whose stability region $A(x_s)$ is not dense in R^n . Let E_1 be the set of equilibrium points and $\hat{c} = \min_{x_i \in \partial A(x_s) \cap E_1} V(x_i)$, then

- [1] $S_{\hat{c}}(x_s) \subset A(x_s)$ and
- [2] the set $\{(S_b(x_s) \cap \bar{A}^c(x_s))\}$ is nonempty for any number $b > \hat{c}$.

The above theorem asserts that the connected component $S_{\hat{c}}(x_s)$ is the best candidate to approximate the stability region $A(x_s)$. Part 2 of Theorem 5.13 asserts that the scheme of choosing $\hat{c} = \min_{x_i \in \partial A(x_s) \cap E_1} V(x_i)$ is optimal because the estimated stability region characterized by the corresponding constant energy surface is the largest one within the entire stability region.

Based on Theorems 5.11–5.13, we propose the following scheme to estimate the stability region $A(x_s)$ of a nonlinear dynamical system (Eq. 5.1) via an energy function $V(\cdot)$.

Scheme: Optimal Estimation of the Stability Region $A(x_s)$ via an Energy Function $V(\cdot)$

A. Determining the critical level value of the energy function

Step A1. Find all the equilibrium points.

Step A2. Order these equilibrium points whose corresponding values, $V(\cdot)$, are greater than $V(x_s)$.

Step A3. Of these, identify the one with the lowest energy function value and whose unstable manifold converges to the SEP x_s . (Let this one be $\hat{x}$.)

Step A4. The value of the energy function at $\hat{x}$ gives the critical level value of this energy function (i.e., $V(\hat{x})$).

B. Estimating the stability region $A(x_s)$

Step B1. The connected component of $\{x | V(x) < V(\hat{x})\}$ containing the SEP x_s gives the estimated stability region.

Remark: The computation associated with Step A1 could be very involved. Efficient numerical methods in conjunction with utilizing special properties of the system under study are needed to implement this step. The above scheme is general and applicable to a nonlinear dynamical system (Eq. 5.1) that has an energy function. The computation efficiency of this scheme, however, can be improved by focusing on type-one equilibrium points if the quasi-stability region is to be estimated. In this situation, instead of finding all of the equilibrium points in Steps A1 and A2, one merely needs to find and check all the type-one equilibrium points.

For the purpose of illustration, we consider the following simple example:

$$\begin{aligned}\dot{x}_1 &= -\sin x_1 - 0.5 \sin(x_1 - x_2) + 0.01 \\ \dot{x}_2 &= -0.5 \sin x_2 - 0.5 \sin(x_2 - x_1) + 0.05.\end{aligned}\tag{5.29}$$

It is easy to show that the following function is an energy function for the system (Eq. 5.29):

$$V(x_1, x_2) = -2 \cos x_1 - \cos x_2 - \cos(x_1 - x_2) - 0.02x_1 - 0.1x_2.\tag{5.30}$$

The point $x^s := (x_1^s, x_2^s) = (0.02801, 0.06403)$ is the SEP whose stability region is to be estimated. Applying the optimal scheme to the system (Eq. 5.29), we have the following:

Steps A1 and A2. There are two type-one equilibrium points within the region $\{(x_1, x_2) : x_1^s - \pi < x_1 < x_1^s + \pi, x_2^s - \pi < x_2 < x_2^s + \pi\}$.

Step A3. The type-one equilibrium point (0.04667, 3.11489) is the one with the lowest energy function value among all the UEPs on the stability boundary $\partial A(x^s)$, and its unstable manifold converges to the SEP (0.02801, 0.06403).

Step A4. The critical level value for the system (Eq. 5.29) is -0.31329 (see Table 5.1).

Curve *A* in Figure 5.2 is the exact stability boundary $\partial A(x^s)$ of the system (Eq. 5.29) according to Corollary 5.3. Curve *B* is the stability boundary estimated by the

Table 5.1 The Coordinates of Two Type-One Equilibrium Points and Their Energy Function Values

Type-one equilibrium point	x_1	x_2	$V(.)$
1	0.04667	3.11489	-0.31329
2	-3.03743	0.33413	2.04547

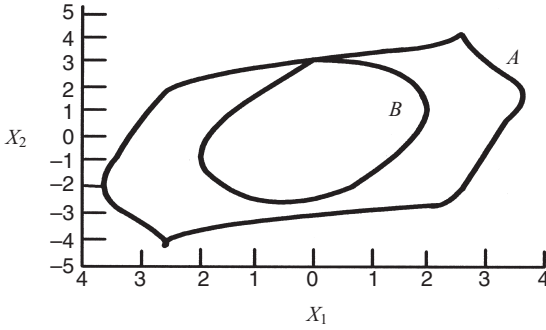


Figure 5.2 Curve A is the exact stability boundary $\partial A(x^s)$ of the system (Eq. 5.29) according to Corollary 5.3. Curve B is the stability boundary estimated by the constant energy surface (with a level value of -0.31329) of the energy function. The optimality of this estimation is also shown in this figure.

connected component (containing the SEP x^s) of the constant energy surface of the energy function (Eq. 5.30) passing through -0.31329 . It can be seen from Figure 5.2 that the critical level value -0.31329 determined by the proposed scheme is indeed the optimal value of the energy function (Eq. 5.30) for estimating the stability region $A(x^s)$. The estimate by curve B , however, is still conservative compared to the exact stability boundary.

5.6 QUASI-STABILITY REGION AND ENERGY FUNCTION

In this section, we will establish a relationship between the structures of a quasi-stability boundary and constant energy surfaces at different level values. Recall that only one component of the constant energy surface can intersect the quasi-stability region. The constant energy surface with a small level value is very conservative in approximating the quasi-stability boundary. As the constant energy surface is expanded with the increasing level value, the approximation becomes more accurate until this constant energy surface intersects the stability boundary at the closest UEP (see Figure 5.3). Two scenarios can occur: the first scenario is that the closest UEP lies in the quasi-stability region and not on the quasi-stability boundary (see Figure 5.3), and the second scenario is that the closest UEP lies on the stability boundary but not on the quasi-stability region.

If the closest UEP does not lie on the quasi-stability boundary, then it must lie on the quasi-stability region. Even as we increase the level value, the new constant energy surface would still lie inside the quasi-stability region $A_p(x_s)$ and it is a connected but not simply connected set, as shown in the following.

Proposition 5.14: Bounded Property

Consider the nonlinear dynamical system described by Equation 5.1, which admits an energy function $V(x)$. Let A , ∂A , and A_p be the stability region, the stability boundary, and the quasi-stability region, respectively, of x_s . Let $c = \min_{x \in \partial A \cap E} V(x)$, where

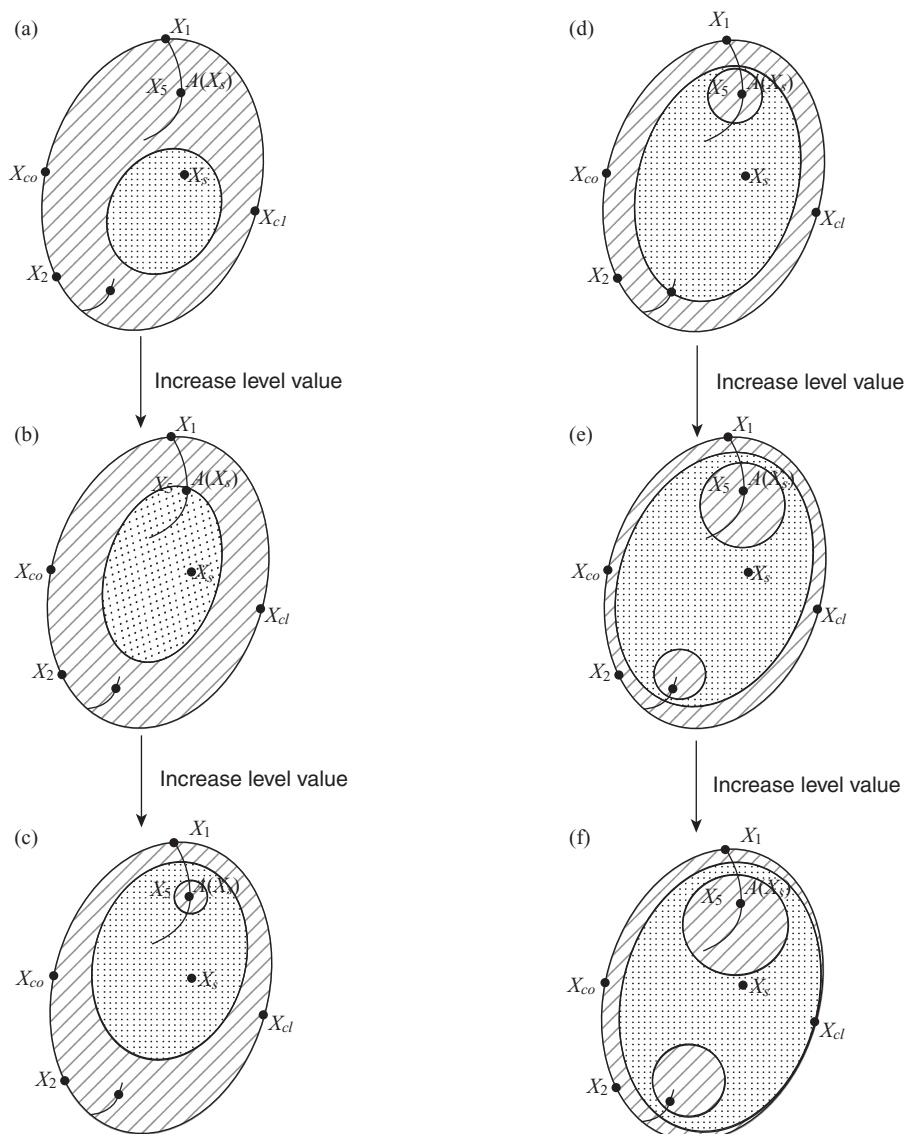


Figure 5.3 The structure of a constant energy surface changes with the increase of its level value. The constant energy surface contains several connected components; some of them are simply connected, while others are not simply connected.* In (a), only one connected constant energy surface intersects with the quasi-stability region and the constant energy surface is conservative in approximating the quasi-stability boundary. In (b), the constant energy surface intersects with the closest UEP, which lies inside the quasi-stability region. In (c), as the level value increases, there is a “hole” in the connected constant energy surface, which is then not simply connected. In (d), the constant energy surface intersects with another UEP, which lies inside the quasi-stability region. In (f), with the increase in the level value, the constant energy surface eventually intersects with a type-one UEP lying on the quasi-stability boundary.

*A set is said to be connected if the line connecting two points in the set lies in the set. A set is said to be simply connected if the set is connected and the (topological) boundary of the set is also connected.

E is the set of equilibrium points of Equation 5.1. If the value c is attained at an equilibrium point $\hat{x} \in A_p$, then for any $\varepsilon > 0$ sufficiently small, the set $\overline{S_{c+\varepsilon}(x_s)}$ is contained in the quasi-stability region; that is, $\overline{S_{c+\varepsilon}(x_s)} \subset A_p$.

Proof: Let k be the type of $\hat{x}$. Since $\hat{x}$ is a hyperbolic equilibrium point, by the Morse lemma (Milnor, 1973), there exists local coordinates $(x_1, \dots, x_k, y_1, \dots, y_{n-k})$ defined in the neighborhood U centered at $\hat{x}$, such that $V(x, y) = c - |x|^2 + |y|^2$ in these coordinates. In these local coordinates, let $H(\varepsilon)$ be a rectangular neighborhood of $\hat{x}$ defined as

$$H(\varepsilon) = \{(x, y) \in U : |y|^2 \leq \varepsilon \text{ and } |x|^2 \leq 2\varepsilon\}.$$

The set H is called a handle of equilibrium point $\hat{x}$. It can be shown that $\overline{S_{c+\varepsilon}(x_s)}$ is homeomorphic to $\overline{S_{c-\varepsilon}(x_s)} \cup H(\varepsilon)$ (Franks, 1980). Note that $\overline{S_{c-\varepsilon}(x_s)}$ and $H(\varepsilon)$ are compact sets. Thus, $\overline{S_{c+\varepsilon}(x_s)}$ is also compact because the image of a compact set under a homeomorphism is compact. Hence, $\overline{S_{c+\varepsilon}(x_s)}$ is bounded.

Now, since $\hat{x} \in A_p = \text{int}\bar{A}$ for ε small enough, it follows that $H(\varepsilon) = \text{int}\bar{A}$. Moreover, $\overline{S_{c-\varepsilon}(x_s)} \subset \text{int}\bar{A}$. This implies that $H(\varepsilon) \cup \overline{S_{c-\varepsilon}(x_s)} = \overline{S_{c+\varepsilon}(x_s)} \subset \text{int}\bar{A}$. This completes the proof.

Proposition 5.14 asserts that if one keeps on increasing the level value from $c + \varepsilon$ to a higher value, say, c' , without hitting $\partial\bar{A}$, then $S_{c'}(x_s)$ is still inside A_p and is bounded. In addition, it can be shown that if $a < b$ are regular values and there are no equilibrium points in $S_b(x_s) - S_a(x_s)$, then $S_a(x_s)$ and $S_b(x_s)$ are diffeomorphic. Hence, the stability region estimated by a suitable constant energy surface always has a nice geometrical structure because it is always homeomorphic to an open n -dimensional sphere. The quasi-stability region estimated by a constant energy surface, however, can exhibit a very complicated shape, as we can see in the following proposition.

Proposition 5.15: Connected but Not Simply Connected Property

Consider the nonlinear dynamical system described by Equation 5.1, which has an energy function $V(x)$. Let A , ∂A , and A_p be the stability region, the stability boundary, and the quasi-stability region, respectively, of x_s . Let $c = \min_{x \in \partial A \cap E} V(x)$, where E is the set of equilibrium points of Equation 5.1. If the value c is attained at an equilibrium point $\hat{x} \in A_p$, then $\overline{S_{c+\varepsilon}(x_s)}$ is not simply connected.

Proof: We will first review the concept of deformation retraction (Massey, 1967). Let B be a subset of X . A deformation retraction of X onto B is a continuous map $F: X \times I \rightarrow X$, where I is the unit interval, such that $F(x, 0) = x$ for $x \in X$, $F(x, 1) \in B$ for $x \in X$, and $F(b, t) = b$ for $b \in B$. If such an F exists, then B is called a deformation retract of X . One can visualize a deformation retraction as a gradual collapsing of the space X onto the subspace B , such that each point of B remains fixed during the shrinking process. This concept is used in the following lemma.

Lemma 5.16:

Let n_u be the type of $\hat{x}$ and let $W_\epsilon^u(\hat{x})$ be an n_u -dimensional neighborhood of $\hat{x}$ in $W^u(\hat{x}) \cap H(\epsilon)$, where $H(\epsilon)$ is the rectangular neighborhood. Then, $S_{c+\epsilon}(x_s)$ has $S_{c-\epsilon}(x_s) \cup W_\epsilon^u(\hat{x})$ as a strong deformation retract.

Lemma 5.16 says that in changing the level value from c to $c + \epsilon$, $S_{c+\epsilon}(x_s)$ can be continuously transformed into $S_{c-\epsilon}(x_s) \cup W_\epsilon^u(\hat{x})$. Therefore, any topological invariant of the first set is preserved after the collapsing.

Now we will prove the proposition. Since $\hat{x} \in A_p = \text{int}\bar{A}$, it follows that $\{W^u(\hat{x}) - \hat{x}\} \subset A$, which implies $W^u(\hat{x}) \subset \text{int}\bar{A}$. Hence, $W_\epsilon^u(\hat{x}) \subset \text{int}\bar{A}$. From Lemma 5.16, it follows that since $S_{c+\epsilon}(x_s)$ has $S_{c-\epsilon}(x_s) \cup W_\epsilon^u(\hat{x})$ as a strong deformation retract, $S_{c+\epsilon}(x_s)$ must self-intersect. If not, then $W_\epsilon^u(\hat{x})$ would connect $S_{c-\epsilon}(x_s)$, which lies in $\text{int}\bar{A}$ to another component $C \subset (\bar{A})^c$ of the level set corresponding to the level value $c - \epsilon$. So before reaching the level value $c - \epsilon$, the component C must intersect $\partial\bar{A}$ and $S_{c-\epsilon}(x_s) \cap C = \emptyset$. This contradicts the fact that, for a given level value, exactly one component of the level set intersects $\bar{A}$, and this component must be $S_{c+\epsilon}(x_s)$. Since $S_{c-\epsilon}(x_s)$ is homeomorphic to an n -dimensional disk, D^n , it can be retracted to a line segment. $W_\epsilon^u(\hat{x})$ would connect $S_{c-\epsilon}(x_s)$, which must self-intersect. So, we conclude that $S_{c-\epsilon}(x_s) \cup W_\epsilon^u(\hat{x})$ can be retracted to a circle. Since the circle is not simply connected and $S_{c+\epsilon}(x_s)$ has the same homotopy type as the circle (Massey, 1967), we conclude that $S_{c+\epsilon}(x_s)$ is not simply connected. This completes the proof.

5.7 CONCLUSION

In this chapter, we have presented a comprehensive energy function theory for general nonlinear autonomous dynamical systems. This comprehensive theory includes the following analytical results:

- global behavior of trajectories,
- global behaviors of trajectories on the stability boundary,
- characterizations of the stability boundary,
- structures of equilibrium points on the stability boundary,
- sufficient conditions for an unbounded stability boundary, and
- distinct energy function values of equilibrium points.

In addition, the theoretical basis for using energy functions to estimate stability regions of large-scale nonlinear systems has been presented. The relationship between the structures of the quasi-stability boundary and the constant energy surfaces at different level values has been established. Optimal schemes for estimating the stability region and quasi-stability regions have been presented, and these schemes form the basis for the closest UEP method. The energy function theory can be applied to power system transient stability models to develop a theoretical foundation for direct methods. We will present this development in later chapters.

One important extension of energy functions is to allow derivatives along system trajectories to be positive in some bounded sets instead of being nonpositive in the state space. We term this extension of energy functions as a generalized energy function. To this end, the extended invariance principle seems promising (Rodrigues et al., 2000, 2001). If the extension is successful, then it is expected that generalized energy functions are applicable to more complicated power system stability models than current transient stability models.